Programming in Mathematics

MATH252 Programming in Mathematics is a college-level course for students enrolled at ISU in the Mathematics or Mathematics Teaching major programs. Students must have a C or better in MATH132 Calculus II to take the course.

Stage 1 – Desired Results

Established Goals

Students will develop and enhance R programs that consists of functions (built-in or created), variables, vectors, matrices, arrays, lists, data frames, and/or graphics.

Students will also construct R programs and apply them to the fields of mathematics, statistics, computer science, physics, and/or related areas.

Note that this plan is for the first part of the course, i.e. the first paragraph. The second part of the course (the second paragraph) will

Established COs

CO #1
Compose basic commands in the R command line

CO #2
Assemble R codes involving student-created functions and R built-in functions

Established MOs

MO 1.1 Compose basic math operations in the R command line
MO 1.2 Create variables
MO 1.3 Derive the output of code that uses Boolean operations
MO 1.4 Create vectors
MO 1.5 Modify elements and subsections of vectors
MO 1.6 Verify the attributes of variables and vectors
MO 1.7 Compare and contrast NA and NULL
MO 1.8 Specify when element recycling occurs

MO 2.1 Create R scripts within R or RStudio
MO 2.2 Compose functions within R that utilize variables and vectors
MO 2.3 Determine arguments for created or R-built-in functions
MO 2.4 Compose functions that within R that utilize loops

Transfer

Students will be able to independently use their learning to…

- View programming as a method of enhancing or speeding up the calculation process, especially in cases where large amounts of data or complex computations are considered
- Seek out programming applications in the fields of mathematics, mathematics education, or related fields.

Meaning

UNDERSTANDINGS

Students will understand that…

- The lessons present one or more methods to apply code to various problems but may not be the only methods.
- Aspects of this programming course may be applied to fields related to mathematics but may not directly be applied to the students' future career fields. Thus, some information may need to be inferred.

ESSENTIAL QUESTIONS

Students will keep considering…

- Why is computer programming important?
- Why is it important to follow a programming language's syntax?
- How will I apply programming in my future work in mathematics or mathematics education?
- Is there another way I can produce the same or similar results in R?

Acquisition of Knowledge and Skill

Students will know…

- R programming terms
- How to read codes written in R
- Structural information about variables, vectors, functions,

Students will be skilled at…

- Entering R commands into the R command line
- Writing code involving variables, vectors, functions,

expand from CO #5		**MO 2.5** Predict the return value(s) of functions	matrices, arrays, lists, and data frames • How to create key objects in R programming	graphics, matrices, arrays, lists, and/or data frames
	CO #3 Create lists in R	**MO 3.1** Create lists in R **MO 3.2** Modify components of lists, with and without tags **MO 3.3** Verify the attributes of lists **MO 3.4** Apply functions to lists **MO 3.5** Combine multiple lists together **MO 3.6** Classify object types in R, including various vector classes and lists **MO 3.7** Create functions for use on lists		
	CO #4 Create matrices and arrays in R	**MO 4.1** Compare and contrast the qualities of vectors, lists, matrices, and arrays in R **MO 4.2** Create matrices and arrays in R **MO 4.3** Modify elements and subsections of a matrix or array **MO 4.4** Verify the attributes of a matrix or array **MO 4.5** Apply mathematical operations on matrices or arrays		

		MO 4.6 Apply functions to matrices and arrays **MO 4.7** Filter elements of matrices and arrays		
	CO #5 Use packages and debug in R	**MO 5.1** Debug programs manually based upon error outputs **MO 5.2** Debug programs by utilizing R functions **MO 5.3** Import packages for use in R		
	CO #6 Create data frames in R	**MO 6.1** Compare and contrast the qualities lists, matrices, and data frames in R **MO 6.2** Create data frames in R **MO 6.3** Modify components data frames **MO 6.4** Verify the attributes of data frames **MO 6.5** Apply functions to data frames **MO 6.6** Combine multiple data frames together		
	CO #7 Write papers in LaTeX	**MO 7.1** Create a LaTeX document **MO 7.2** Create a sections and theorems in LaTeX		

		MO 7.3 Write comments and format a LaTeX document, such as changing font styles, entering lines of code, and changing colors **MO 7.4** Use math mode in LaTeX **MO 7.5** Create a bibliography and image figures in LaTeX **MO 7.6** Apply page formatting, such as adjusting margins and headers in LaTeX		

Stage 2 - Evidence		
Code: **List the COS and MOS with which the assessment aligns** **NOTE: each CO and MO may be listed multiple times**	**Evaluative Criteria**	***Students will show their learning by…***
CO 1 and all its Module-level objectives	• High: code produces expected output • Average: code has minor errors, such as inappropriate use of assignment in R (<- as opposed to =) or other syntax issues • Low: code has major errors or is hard-coded	**Performance Tasks** *Students will show that they really understand by evidence of…* • **Participation 1: leading towards MO 1.1-1.8** ○ Verify the value(s) stored in an object ○ Determine the cause of object error (e.g. notice case-sensitivity) ○ Determine the use of scripts in R ○ Verify the comment symbol in R ○ Explain float arithmetic and issues with this arithmetic ○ Create a vector in R using a variable ○ Given R code, explain the Boolean outputs that occur ○ Given R code, explain element recycling, with and without errors

		• **Homework 1: MO 1.1–1.8** ○ Compute expressions and assign the output to a variable. Do for multiple expressions and variables. State what the output of the variables are. ○ Compute operations on created variables and state what the output of those operations are. ○ Apply Boolean operations with the defined variables and state the output ○ Assign the variables to a vector in a specific order and state the output. ○ Create vectors for use within the assignment ○ Adjust specific values within a vector to different values, including replacing values NA ○ Utilize object recycling, including comparing values with values inside of a vector ○ Consider applications of NA and NULL
		Other Evidence
		Students will show that they have achieved these Stage 1 Goals by… • **Quiz 1: CO 1** ○ **Computer Quiz – students will be permitted to create their programs within the R GUI or RStudio. They are not permitted to use outside resources without permission** ○ Determine the output of simplifying expressions with multiple operations. ○ Give the input for code assigning given values to a vector. ○ Perform operations with given vectors ○ Compare a vector with a number (which utilizes recycling) ○ Identify value locations within a vector ○ Apply Boolean operations

CO 2 and all its Module-level objectives This expands from CO 1	• High: code produces expected output • Average: code has minor errors, such as missing { } or other syntax issues • Low: code has major errors or is hard-coded	**Performance Tasks** *Students will show that they really understand by evidence of…* • **Homework 2: MO 2.1–2.3** ○ Create a function that performs a specific task ○ Create a function that utilizes the use of conditions ○ Use R built-in functions, which require defining arguments, to perform specific tasks • **Homework 3: MO 2.4–2.5** ○ Create a function that performs a specific task that involves loops ○ Adapt a function previously created to utilize loops ○ Given a function, predict the output when given an input for the function. This will include functions that give error outputs **Other Evidence** *Students will show that they have achieved these Stage 1 Goals by…* • **Quiz 2: MO 2.1–2.3** ○ **Computer Quiz – students will be permitted to create their programs within the R GUI or RStudio. They are not permitted to use outside resources without permission** ○ Given a function, identify the parameters ○ Given a function with multiple conditions, determine what conditions need to be met in order to trigger a specific event ○ Given a function with multiple conditions, explain the outputs from multiple function calls. ○ Create a basic function (like “Hello World!”) ○ Explain the error that occurs when trying to create a function with a bad name ○ Explain the error when attempting to use an undefined object

		• **Quiz 3: MO 2.4–2.5** ○ **Computer Quiz – students will be permitted to create their programs within the R GUI or RStudio. They are not permitted to use outside resources without permission** ○ Given a function containing a loop, describe what operations occur and given a sample function call and output, explain the reason for the output ○ Given a function containing an infinite loop, explain why the code has an issue and how to fix the issue to give the desired results
CO 3 and all its Module-level objectives This expands from COs 1 and 2	• High: code produces expected output • Average: code has minor errors, such as missing ( ) or other syntax issues • Low: code has major errors or is hard-coded	**Performance Tasks** *Students will show that they really understand by evidence of…* • **Homework 4: MO 3.6 and MO 3.7** ○ Create a function that utilizes indexing to print items in a list ○ Create a function that counts the different object types in a list • **Homework 5: MO 3.1-3.3 and MO 3.5-3.6** ○ Create a list with various object types as items ○ Utilize list indexing to modify items, name item tags, and perform various operations on items ○ Create a list with specific items from another list ○ Nest a list as an item within another list ○ Compare lapply(), sapply(), and apply() functions ○ Determine the class of an object **Other Evidence** *Students will show that they have achieved these Stage 1 Goals by…* • **Quiz 4: CO 3** ○ **Computer Quiz – students will be permitted to create their programs within the R GUI or RStudio. They are**

		not permitted to use outside resources without permission ○ Determine the object type of an item in a list ○ Determine from a list of options which methods of indexing provide an expected output ○ Determine from a list of options which method of indexing creates a new item in a list given specific standards, such as object type and item tag ○ Determine from a list of options which option creates a new list from a previous list meeting specific standards ○ Given how an R object is defined, explain the reason the object is assigned a specific class
CO 4 and all its Module-level objectives This expands from COs 1 and 2	• High: code produces expected output • Average: code has minor errors, such as missing ( ) or other syntax issues • Low: code has major errors or is hard-coded	**Performance Tasks** *Students will show that they really understand by evidence of…* • **Homework 6: MO 4.2–4.7** ○ Create matrices in R ○ Confirm matrix dimensions given removal of rows or columns in R ○ Perform given matrix operations in R (basic operations) ○ Explain error outputs from given matrix operations ○ Compare standard multiplication and matrix multiplication • **Homework 7: MO 4.1, 4.2, and 4.6** ○ Compare R operations on matrices that appear similar, but output different object types (e.g. vector vs. matrix) ○ Store vectors as matrices in R ○ Perform given matrix operations in R (matrix operations) **Other Evidence** *Students will show that they have achieved these Stage 1 Goals by…* • **Quiz 5: CO 4** ○ **Computer Quiz – students will be permitted to create their programs within the R GUI or RStudio. They are**

		not permitted to use outside resources without permission ○ Create a matrix and an in R ○ Alter the values within a matrix ○ State the R code that verifies specific attributes of matrices ○ Operations on matrices ○ Filter elements of matrices ○ Evaluate a given function, which performs an operation on a matrix, and determine its output
CO 5 and all its Module-level objectives	• High: code produces expected output • Average: code has minor errors, such as missing ( ) or other syntax issues • Low: code has major errors or is hard-coded	**Performance Tasks** *Students will show that they really understand by evidence of…* • **Homework 8: MO 5.1-5.2** ○ Given an R script, debug the program • **Participation 3: MO** ○ Import various R packages based upon provided instructions **Other Evidence** *Students will show that they have achieved these Stage 1 Goals by…* • **Final Project** ○ **Group activity – students will have access to computers, notes, and will collaborate with 1-2 other classmates to complete a provided task** ○ Debug issues within created code ○ **(Optional)** Import packages for use with project • **Participation 4** ○ **Group activity – students will have access to computers, notes, and will collaborate with 1-2 other classmates to complete a provided task** ○ Students will need to import packages to perform various computational tasks

CO 6 and all its Module-level objectives Expands from the previous COs	• High: code produces expected output • Average: code has minor errors, such as missing ( ) or other syntax issues • Low: code has major errors or is hard-coded	**Performance Tasks** *Students will show that they really understand by evidence of…* • **Participation 4 – MO 5.3, 6.2, 6.4-6.5** ○ **Group activity – students will have access to computers, notes, and will collaborate with 1-2 other classmates to complete a provided task** ○ Import various packages to import a data frame and graphical tools into R ○ Perform column operations on the provided data frame ○ Apply functions to the provided data frame ○ Compare various function outputs from applying functions to the provided data frame • **Homework 9: MO 6.1-6.4, 6.6** ○ Create a data frame in R ○ Append a column to a data frame in R ○ Rearrange columns of a data frame in R ○ Combine data frames in R ○ Modify items on a data frame in R ○ Filter items of a data frame **Other Evidence** *Students will show that they have achieved these Stage 1 Goals by…* • **Quiz 6: CO 6** ○ **Computer Quiz – students will be permitted to create their programs within the R GUI or RStudio. They are not permitted to use outside resources without permission** ○ Create a data frame ○ Given a list of options, determine which combines two data frames based upon columns ○ Given a list of options, determine which combines two data frames based upon rows ○ Modify an item in a data frame ○ Verify the attributes of a given data frame ○ Verify various object types

CO 7 and all its Module-level objectives	• High: code produces expected output, with or without minor errors • Average: code does not produce expected output, such as not appropriately labelling indexed items • Low: code has major errors that does not allow for a PDF to be compiled	**Performance Tasks** *Students will show that they really understand by evidence of…* • **Participation 5** ○ Create a LaTeX document ○ Create sections, including abstract, introduction, literature review, methods, results, and conclusion ○ Use math mode • **Homework 10: MO 7.1–7.6** ○ Create a LaTeX document, including a title and author ○ Format the document with specific margins, font, font sizes, and colors ○ Write comments in LaTeX code ○ Create sections and theorems, including proofs ○ Create math equations ○ Insert an image ○ Compile a LaTeX document into a PDF **Other Evidence** *Students will show that they have achieved these Stage 1 Goals by…* • **Final Project** ○ **Group activity – students will have access to computers, notes, and will collaborate with 1-2 other classmates to complete a provided task** ○ Write a paper in LaTeX outlining the project, including examples of code, explanation of code, and explanation of mathematical knowledge necessary for the code

Stage 3 – Learning Plan

Summary of Key Learning Events and Instruction

They key to the transfer and meaning goals is that math and math education majors need to understand programming and how to apply it to their respective fields. The aim is for students to be able to utilize programming in a manner that assists with future computations in mathematics or for utilizing programming in mathematics teaching (depending on each student's degree path). Though it is not outlined above, the delivery of this course will also serve to encourage group discussions in a productive environment to benefit the students as they continue the pursuit of their degrees in future math courses. These goals will guide them into Phase 2 of the course (which will be planned separately from this project).

Unless otherwise stated, students will be able (and are generally expected to utilize) the R GUI or RStudio (or another teacher-approved software) along with online resources.

- Teacher supplies instructions on how to install the R GUI as well as RStudio. The teacher will also supply information regarding viable alternatives, should any exist.
- Teacher supplies instructions on how to download digital instructions, should this be necessary.
- Via Canvas, SoftChalk, Swirl/Swirlify, Learn, RMarkdown/bookdown, or other open resources, the teacher will provide students with flipped-instruction for students to review prior to class meetings, predetermined by the teacher.